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# YOLO-CMDS: An enhanced method for PCB defect detection

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## ABSTRACT

Surface defects in printed circuit boards (PCBs) compromise product quality, which in turn undermines equipment stability and reliability. Detecting small defects in PCBs is challenging due to their compact layout and complex background. This paper introduces a novel YOLOv8-CGB\_Down-MLCA-DPFHead-Slim-Neck (YOLO-CMDS) network, an enhanced version of YOLOv8, designed to improve the accuracy of small target detection. Key enhancements include the CGB\_Down module, which improves feature extraction for small defects in complex backgrounds; the integration of a mixed local channel attention mechanism (MLCA), which boosts detection accuracy with minimal extra parameters; the addition of the DPFHead detection head, further improving the model's capability to identify small defects in PCB detection tasks; the Slim-Neck module facilitates feature fusion between the backbone network and detection head, improving the model's ability to detect minute defects and increasing overall accuracy. Extensive testing on a public PCB dataset shows that YOLO-CMDS achieves a 99.13% accuracy rate in detecting small targets and maintaining accuracy balance in complex scenarios, outperforming several leading algorithms. These results demonstrate that YOLO-CMDS is a reliable and competitive solution for PCB defect detection.

## 1. Introduction

Detecting surface defects is essential for ensuring the quality of printed circuit boards (PCBs) [1–3]. Traditional PCB inspection relies on optical equipment for detecting common defects. The growing demand for PCBs in electronics and emerging technologies has driven manufacturing toward higher reliability, density, and miniaturization [4,5]. Compact layouts, higher line densities, and smaller components [6,7] challenge the accuracy of traditional inspection methods. Key challenges include:

1. The intricate PCB manufacturing process is prone to defects like open circuits, short circuits, and burrs [8–10]. Defective PCBs can cause leakage or fires, posing safety risks. This raises challenges for models to detect defects efficiently and accurately.
2. PCB surface defects are typically small due to manufacturing constraints. In high-resolution PCB images, defects often occupy less than 0.07% of the image [11]. Detecting such small defects quickly and accurately is challenging.

Data-driven approaches leveraging machine vision and deep learning have gained prominence in PCB defect detection [12–17]. Deep

learning methods for PCB defect detection are typically classified as classification, object detection, or segmentation networks [18]. Object detection networks include two-stage methods like Faster R-CNN, which generate region proposals, and one-stage methods like YOLO, which predict class and location in a single pass. For example, Hu and Wang [19] used Faster R-CNN with ResNet50 to detect small PCB defects. Liu and Wen [20] introduced MobileNet-YOLO-Fast for real-time PCB defect detection. Li et al. [21] combined Hybrid-YOLOv2 and Faster R-CNN for detecting DIP soldering defects. Despite progress, these methods struggle with real-time and accurate small defect detection in complex PCB backgrounds.

To address the above challenges, this paper proposes YOLO-CMDS, an enhanced YOLOv8-based framework specifically designed for accurate and efficient detection of small PCB defects in complex industrial environments. Unlike conventional methods that rely on a single optimization strategy, YOLO-CMDS improves small-defect detection performance through a joint optimization of feature extraction, feature enhancement, and feature fusion. By integrating contextual information, lightweight attention mechanisms, and an optimized detection head, the proposed model effectively balances detection accuracy and computational efficiency, making it suitable for practical PCB inspection tasks.

The main contributions of this paper can be summarized as follows:

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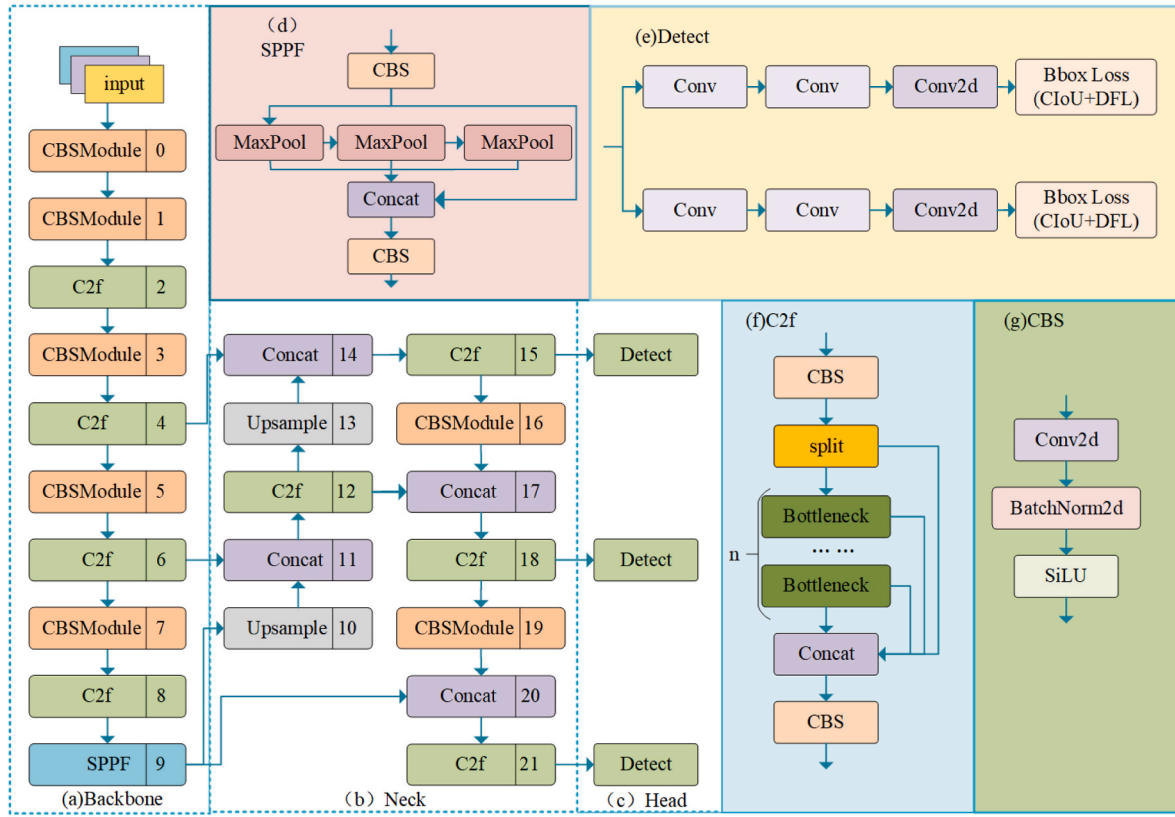


Fig. 1. Framework of YOLOv8.

- (1) We propose a context-enhanced feature extraction strategy for PCB defect detection by introducing the CGB\_Down module into the backbone network. This module jointly models local, surrounding, and global contextual information, significantly improving the representation of small defects under complex backgrounds.
- (2) A lightweight Mixed Local Channel Attention (MLCA) mechanism is integrated into the neck to enhance discriminative feature learning with minimal additional parameters. This design effectively strengthens the model's focus on defect regions while maintaining computational efficiency.
- (3) An optimized detection architecture is constructed by combining the DPFHead and Slim-Neck modules, enabling efficient multi-scale feature fusion and improved small-target detection capability. This design further enhances detection robustness without introducing excessive computational overhead.

These improvements enable YOLO-CMDS to achieve superior detection accuracy for small PCB defects while preserving real-time performance, demonstrating its effectiveness and practicality for industrial PCB inspection applications.

## 2. Related work

This section reviews traditional PCB defect detection methods and introduces the YOLOv8 model, explaining its background and design.

### 2.1. Methods for PCB surface defect detection

Before machine vision, PCB defect detection primarily relied on manual visual inspection. Some companies still rely on manual inspection for quality control. Manual inspection is inefficient and prone to errors, making it unsuitable for modern production needs [18].

Automated optical inspection (AOI) has emerged as a key tool for detecting PCB surface defects using optical principles. Raihan et al. [22] used OpenCV and image subtraction for rapid PCB defect detection. Kim et al. [23] developed a convolutional autoencoder to locate defects by comparing reconstructed and original images. Li and Qu et al. [24] used feature pyramid networks to enhance small target detection. Yang and Li [25] implemented an AOI-based system to detect open circuits, short circuits, and scratches. AOI performs well in specific cases but struggles with the growing complexity of industrial manufacturing. Specialized schemes are needed to handle issues like large-area shadows. The high cost of AOI systems restricts their adoption by smaller enterprises.

Traditional methods like image segmentation and feature extraction are also widely used. Techniques like thresholding, edge detection, and region segmentation are common. Cai et al. [26] used an improved ViBe model for PCB defect detection. Ardhy et al. [27] introduced adaptive Gaussian thresholding for a low-cost vision system. Traditional methods extract features like color, texture, shape, and statistics. Zhang et al. [28] used SIFT features but noted room for accuracy improvement. Li et al. [29] applied edge gradient-based feature extraction to reduce light interference and improve detection speed.

Advances in deep learning have driven progress in PCB defect detection. Deep learning object detection includes two-stage methods (e.g., Faster R-CNN) and single-stage methods (e.g., YOLO and SSD) [19, 20, 30]. Hu et al. [19] used Faster R-CNN with image pyramids to capture features and reduce false negatives. Single-stage methods are generally more efficient. Li et al. [21] leveraged YOLOv2 to enhance detection speed.

### 2.2. YOLOv8 model

YOLOv8, developed by Ultralytics, is a widely used model in the YOLO object detection series. It comprises four components: input layer, backbone, neck, and output layer.

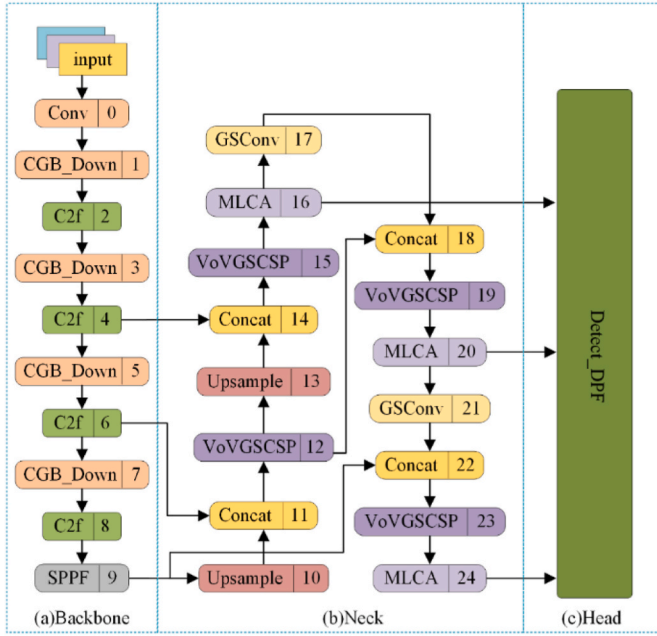


Fig. 2. Framework of YOLO-CMDS

Fig. 1 illustrates the YOLOv8 architecture, highlighting the backbone and neck modules. The backbone extracts multi-scale features using convolutional (Conv) and C2f units. As shown in Fig. 1, the backbone of YOLOv8 is responsible for extracting multi-scale features from the input image using a series of convolutional building blocks. Specifically, the CBSModule refers to a standard convolutional block composed of a Convolution layer, a Batch Normalization layer, and a SiLU activation function, which is widely used for basic feature extraction. The C2f module is an improved variant of the C3 module, designed to reduce computational complexity while enhancing gradient flow by adopting a more efficient feature reuse strategy. By combining CBSModule and C2f units, the YOLOv8 backbone effectively balances feature representation capability and computational efficiency. The extracted features are further processed by the Spatial Pyramid Pooling Fast (SPPF) module before being forwarded to the neck network for multi-scale feature fusion. C2f, an improved C3 unit, reduces convolutional layers and enhances gradient flow. The Spatial Pyramid Pooling (SPPF) module processes features before passing them to the neck. The neck combines Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) to fuse high- and low-level features, enhancing semantics and localization.

YOLOv8 separates the classification and detection heads and employs TaskAlignedAssigner [29]. This method uses classification and regression scores to assign samples, improving accuracy. The loss

function includes binary cross-entropy (BCE) for classification and Dynamic Focal Loss (DFL) with Center IoU (CIoU) for regression.

Overall, YOLOv8 excels as a classic model in the YOLO series. However, it faces challenges in PCB defect detection. Enhancing detection accuracy and speed is essential for industrial PCB inspection.

### 3. YOLO-CMDS model

#### 3.1. Overall framework

Fig. 2 illustrates the YOLO-CMDS model, comprising a feature extraction backbone, feature fusion neck, and output detection head. YOLO-CMDS builds on and optimizes the YOLOv8 framework. The backbone integrates the CGB\_Down [31] module to boost feature extraction and detection accuracy. The neck uses a Mixed Local Channel Attention (MLCA) [32] strategy, enhancing multi-scale defect detection with minimal added parameters. The DPFHead detection head mitigates class imbalance and improves small-object detection. The Slim-Neck [33] module links the backbone and detection head. It facilitates feature fusion and processing, increasing robustness and detection accuracy.

#### 3.2. Feature extraction method based on CGB\_Down

PCB defect detection is challenging due to many defect-free components and low contrast between target areas and the background, which hinders accurate feature extraction.

To address this, the CGB\_Down module replaces traditional convolutional units, improving small target defect detection. CGB\_Down boosts feature extraction and accuracy by leveraging local, contextual, and global image information. It focuses on utilizing contextual relationships to help the model interpret image content more accurately.

As shown in Fig. 3, the CGB\_Down module has four submodules: (1) Local Feature Extractor (floc) uses standard convolution to capture critical local details; (2) Surrounding Context Extractor (fsur) applies dilated convolution to expand the receptive field and gather contextual data; (3) Joint Feature Extractor (fjoi) merges local and contextual features for a richer representation; and (4) Global Context Extractor (fglo) employs global average pooling and a multilayer perceptron to emphasize key features, enhancing detection accuracy and efficiency.

#### 3.3. Attention mechanism based on MLCA

Accurately detecting defects in PCB images is challenging due to the difficulty in differentiating defect areas from the background. To overcome this, we incorporate the lightweight MLCA (Mixed Local Channel Attention) module to enhance focus on critical regions. As shown in Fig. 4, MLCA combines channel-level and spatial-level information, balancing local details and global context. Its key strength is improving recognition capabilities with minimal computational overhead. It is

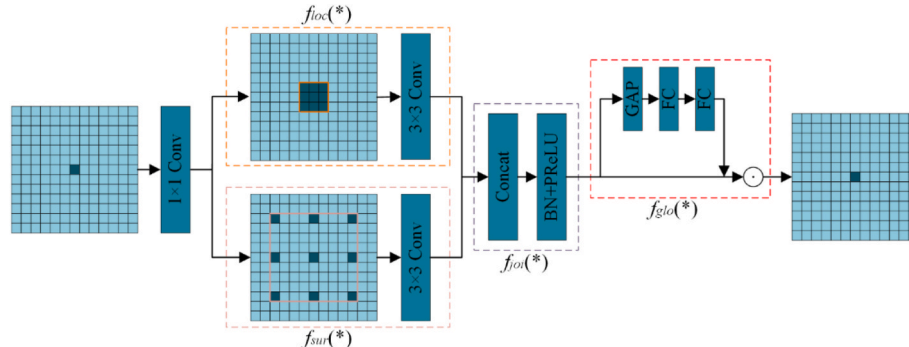


Fig. 3. Structure of CGB\_Down.

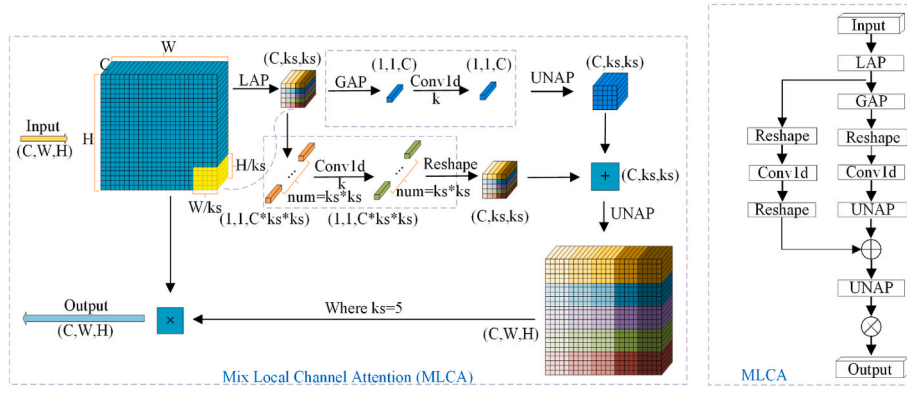


Fig. 4. Structure of MLCA.

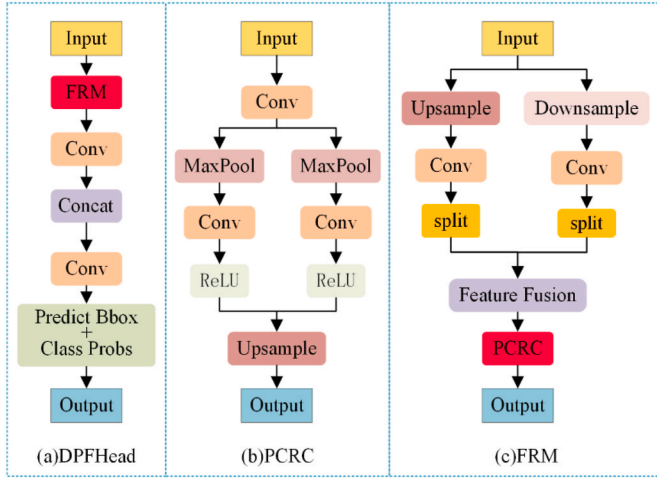


Fig. 5. Schematic diagram of the improved DPFHead and its main modules.

more computationally efficient than YOLOv8's attention mechanisms.

The MLCA module is specifically tailored for feature extraction and fusion, boosting object recognition performance. It extracts fine-grained local features using Local Average Pooling (LAP) and combines them with global features via Global Average Pooling (GAP). This approach forms the basis for feature extraction, represented by the following equation:

$$LAP(x) = \frac{1}{N} \sum_{i=1}^N x_i \quad (1)$$

Where  $LAP(x)$  denotes the local average pooling output of feature  $x$ ;  $N$  is the number of elements in the local region;  $x_i$  is the  $i$ -th element in this region.

$$GAP(x) = \frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H x_{ij} \quad (2)$$

Where  $GAP(x)$  denotes the global average pooling output of feature  $x$ ;  $W$  and  $H$  are the width and height of the feature map  $x$ ;  $x_{ij}$  is the element at position  $(i, j)$  of  $x$ .

$$y(k) = \sum_{i=1}^C w_i \cdot x_i(k) + b \quad (3)$$

Where  $y(k)$  is the feature transformation output at the  $k$ -th step;  $C$  is the number of input feature channels;  $w_i$ ,  $x_i(k)$ , and  $b$  represent the weight of the  $i$ -th channel, the  $i$ -th input feature channel at step  $k$ , and the bias

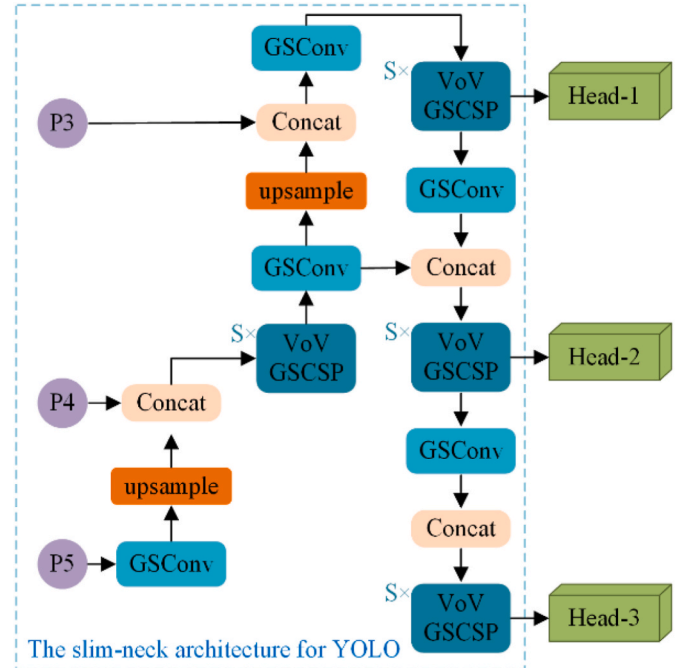


Fig. 6. Structure of slim-neck.

term, respectively.

### 3.4. DPFHead detection head

To improve small defect detection, this study introduces a novel detection head, DPFHead. DPFHead includes Distribution Focus Loss (DFL), the Multi-Scale Feature Fusion Layer (PCRC), and the Feature Reorganization Module (FRM). DFL improves deep feature learning for complex patterns, PCRC integrates multi-scale contextual information for better fusion, and FRM enhances saliency while reducing redundancy. Fig. 5 illustrates its structure.

The FRM module optimizes feature reconstruction and fusion, improving interaction and expressiveness across scales. DFL, a key component, analyzes and processes complex feature maps. PCRC refines input features through a dedicated branching structure. A stacked convolutional structure is adopted, expressed mathematically as:

$$F' = \text{ReLU}(W * F + b) \quad (4)$$

In this formula,  $F'$  is the enriched feature map,  $W$  is the convolution kernel's weight matrix,  $b$  is the bias term, and  $F$  is the input feature map. This method efficiently handles upsampling and downsampling,

**Table 1**  
Training parameters.

| Parameters           | Initialization                     |
|----------------------|------------------------------------|
| learning rate        | 0.001                              |
| Weight_decay         | 0.00005                            |
| number of iterations | 300                                |
| batch size           | 4                                  |
| Label_smoothing      | 0                                  |
| optimizer            | SGD with momentum (factor = 0.937) |

allowing the model to refine multi-scale information.

### 3.5. Sim-neck optimization method for minor defects in PCB

In PCB surface defect detection, minor defects usually occupy only a small number of pixels and are easily overwhelmed by complex background textures. Although the original YOLOv8 neck structure enables multi-scale feature fusion, it introduces redundant feature channels and unnecessary computational overhead when applied to small defect detection tasks. Small targets are generally defined in two ways: by physical dimensions or pixel size, typically smaller than  $32 \times 32$  [34]. Literature [35] indicates that PCB datasets contain numerous small targets, some smaller than  $10 \times 10$  pixels.

To address this issue, we propose a lightweight and efficient Slim-Neck module that connects the backbone and detection head, as illustrated in Fig. 6. The Slim-Neck is designed to streamline feature fusion by reducing channel redundancy while preserving essential spatial and semantic information. Specifically, the module adopts a compact channel configuration and simplified fusion paths, which effectively suppress irrelevant background responses and enhance the representation of small-scale defects.

Unlike conventional neck structures that rely on heavy feature aggregation, the Slim-Neck focuses on task-oriented feature refinement. By maintaining high-resolution feature information and eliminating excessive intermediate operations, the proposed module improves small defect sensitivity without increasing computational complexity. This design enables the network to achieve a better balance between accuracy and efficiency, making it particularly suitable for compact PCB layouts with dense and tiny defect patterns.

These optimizations improve feature fusion and processing while enabling efficient and accurate small target detection through its flexible design.

## 4. Experiments and discussions

This section evaluates the proposed network model on the HRIPCB dataset and compares its performance with other object detection algorithms. Multiple metrics are used for evaluation, with details on the experimental environment and methods provided in later sections.

### 4.1. Experimental design

#### 4.1.1. Experimental environment and training strategies for the model

To validate the RTDETR-DRDA model, we compared it with several

mainstream algorithms, including Faster R-CNN, SSD [36], YOLOv3 [37], YOLOv5, YOLOv8 [38], CDI-YOLO [39], and Light-PDD [1]. The parameters for these algorithms were adjusted based on relevant literature. The model's initialization settings are provided in Table 1.

To ensure fairness, all algorithms were tested on the same dataset with a uniform input image size of  $640 \times 640$ . Training and testing were conducted using the same deep learning framework. The experiment was conducted using PyTorch 2.1.1 and Python 3.8. The hardware used includes an Intel Core i5-12600KF CPU (3.51 GHz), 32 GB RAM, an NVIDIA GeForce GTX 3060 GPU (12 GB VRAM), and Windows 10.

#### 4.1.2. Datasets

The dataset used in this experiment is the HRIPCB public dataset from Peking University, shown in Fig. 7. The dataset consists of 693 PCB defect images, which were augmented to produce 10,668 images. It includes six defect types: missing holes, mouse bites, open circuits, short circuits, spurious copper, and false copper. The dataset is split into training (7467 images), validation (1067 images), and test sets (2134 images) in a 7:1:2 ratio.

#### 4.1.3. Evaluation metrics

This study evaluates the detection performance of the model using multiple classical metrics, including Average Precision (AP) and F1 score. The Precision-Recall (P-R) curve is plotted, with Recall and Precision on the x-axis and y-axis, respectively. The area enclosed by the curve and the axes represents the AP value, which is used to measure the model's detection accuracy for a specific class. The formulas for Precision, Recall, and AP are as follows:

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$AP = \int_0^{Recall} P(r) dr \quad (7)$$

Here, TP represents true positives; FP represents false positives; FN represents false negatives. The P-R curve is generated by the relationship between Recall and Precision, and the AP value is calculated using interpolation.

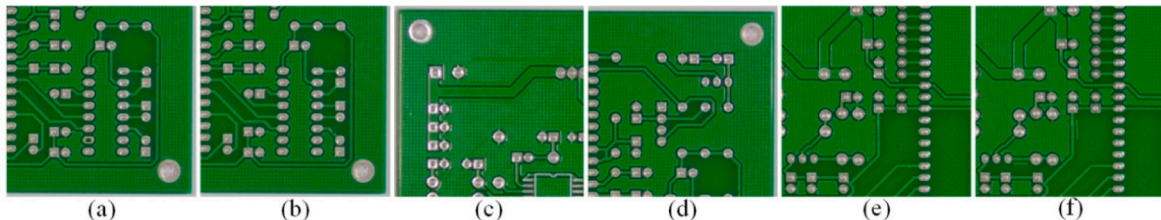
Since PCB defect detection is a multi-class task, this study also reports the mean Average Precision (mAP) for each class, with the following formula:

$$mAP = \frac{\sum_{i=1}^n AP_i}{n} \quad (8)$$

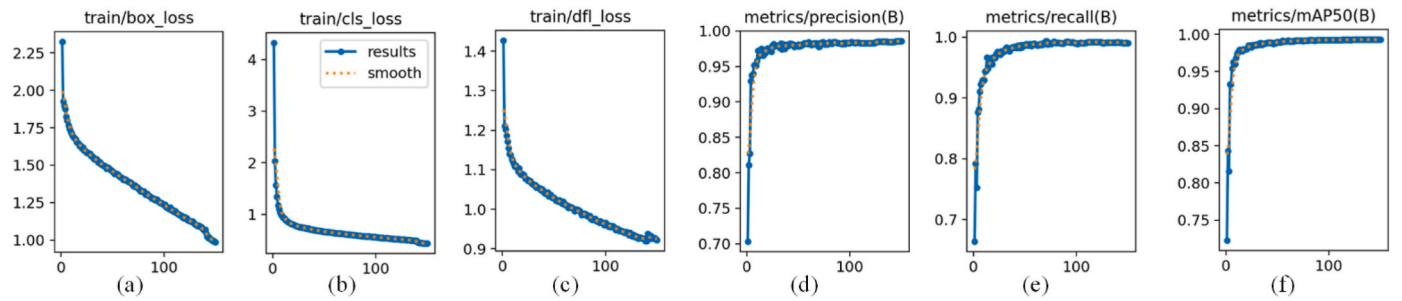
Here,  $n$  is the total number of classes.

The formula for the F1 score is as follows. A higher F1 score indicates stronger generalization ability of the model.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (9)$$



**Fig. 7.** HRIPCB dataset defect diagram: (a)missing hole (b)mouse bite (c)open circuit (d)short (e)spur (f)spurious copper.



**Fig. 8.** Shows the training results of our model on the HRIPCB dataset: (a)-(c) changes in the loss during training, (d) and (e) the model's recall and precision, and (f) the model's accuracy.

**Table 2**

Specific value of defect map.

| Class of Defect | mAP0.5 (%) | mAP0.5-0.95 (%) |
|-----------------|------------|-----------------|
| missing hole    | 99.1       | 69.6            |
| mouse bite      | 99.4       | 67.9            |
| open circuit    | 99.2       | 64.6            |
| short           | 99.1       | 65.7            |
| spur            | 99.1       | 65.5            |
| spurious copper | 98.9       | 65.3            |

**Table 3**

Performance comparison table of model test results.

| Method        | Parameters(million) | mAP0.5 | mAP0.5-0.95 | FPS   |
|---------------|---------------------|--------|-------------|-------|
| Faster R-CNN  | 23.59               | 72.76  | 39.3        | 12    |
| SSD [39]      | 26.9                | 75.3   | 21.4        | 14    |
| YOLOv3 [37]   | 58.7                | 88.36  | 46.6        | 55    |
| YOLOv5        | 6.68                | 94.2   | 48.5        | 112   |
| YOLOv8 [38]   | 30.13               | 98.73  | 58.9        | 78.7  |
| CDI-YOLO [39] | 5.76                | 98.3   | 51.1        | 128   |
| Light-PDD [1] | 8.2                 | 93.8   | 47.1        | 68.71 |
| RT-DETR       | 19.88               | 98.8   | 64.7        | 64.93 |
| Ours          | 32.55               | 99.13  | 66.4        | 68.66 |

#### 4.2. Results and analysis

The model was trained on the HRIPCB dataset, with results shown in Fig. 8. The model converged within 150 epochs, which was set as the training period for this study. Results show a rapid decrease in loss within the first 50 epochs, with training and validation losses closely aligned and showing similar trends. Afterward, the loss continued to decline, while precision, recall, and mAP\_0.5 stabilized between 50 and 150 epochs, achieving optimal training results.

Table 2 shows the training results on both the custom and HRIPCB datasets. For larger defects, like "missing hole" and "spurious copper," detection accuracy exceeds 99%, while smaller defects, such as "mouse bite," "spur," and "short," achieve over 98% accuracy.

To validate the model's effectiveness, we tested it on the HRIPCB dataset and compared its performance with other advanced models. Comparison results are shown in Table 3. Compared to algorithms like RT-DETR and YOLOv8, the proposed model shows significant improvements in mAP\_0.5 and mAP\_0.5:0.95. Additionally, model parameters were compressed, resulting in faster convergence during training. This optimization supports storage and computation efficiency for PCB defect detection on embedded devices, while also enhancing accuracy.

#### 4.3. Ablation experiments

This section conducts ablation experiments to validate the impact of the proposed strategies on model performance. The models used are as

**Table 4**

Ablation experimental results.

|     | Category        | AP/<br>% | Recall/<br>% | F1<br>Score/% | GFLOPs | FPS  | mAP<br>(%) |
|-----|-----------------|----------|--------------|---------------|--------|------|------------|
| (1) | missing hole    | 99.3     | 99.3         | 98.89         | 8.2    | 78.7 | 98.73      |
|     | mouse bite      | 98.5     | 98.1         | 97.54         |        |      |            |
|     | open circuit    | 98.8     | 97.3         | 97.64         |        |      |            |
|     | short           | 98.5     | 96.6         | 97.19         |        |      |            |
|     | spur            | 98.7     | 95.5         | 96.63         |        |      |            |
|     | spurious copper | 98.6     | 97.4         | 96.69         |        |      |            |
| (2) | missing hole    | 99.2     | 99.9         | 99.2          | 8.5    | 76.4 | 98.65      |
|     | mouse bite      | 99       | 98.1         | 98.24         |        |      |            |
|     | open circuit    | 98.9     | 98.1         | 98.44         |        |      |            |
|     | short           | 97.6     | 97.4         | 97.24         |        |      |            |
|     | spur            | 98.7     | 94.8         | 96.45         |        |      |            |
|     | spurious copper | 98.5     | 97.4         | 97.64         |        |      |            |
| (3) | missing hole    | 98.8     | 98           | 98.09         | 8.6    | 75.9 | 98.8       |
|     | mouse bite      | 99.2     | 98.8         | 98.39         |        |      |            |
|     | open circuit    | 98.9     | 98           | 97.84         |        |      |            |
|     | short           | 98.3     | 97.7         | 97.7          |        |      |            |
|     | spur            | 98.7     | 96           | 97.16         |        |      |            |
|     | spurious copper | 98.9     | 97.8         | 97.29         |        |      |            |
| (4) | missing hole    | 99.1     | 99.7         | 99.24         | 8.9    | 72.1 | 99         |
|     | mouse bite      | 99.2     | 98           | 98.39         |        |      |            |
|     | open circuit    | 99.1     | 98           | 98.19         |        |      |            |
|     | short           | 99       | 98           | 97.59         |        |      |            |
|     | spur            | 98.8     | 96.2         | 97.04         |        |      |            |
|     | spurious copper | 98.8     | 97.8         | 97.19         |        |      |            |
| (5) | missing hole    | 99.1     | 99.9         | 99.39         | 9.1    | 68.7 | 99.13      |
|     | mouse bite      | 99.4     | 99.4         | 99.19         |        |      |            |
|     | open circuit    | 99.2     | 99.4         | 98.94         |        |      |            |
|     | short           | 99.1     | 98.6         | 98.39         |        |      |            |
|     | spur            | 99.1     | 97.2         | 98.04         |        |      |            |
|     | spurious copper | 98.9     | 99.3         | 98.69         |        |      |            |

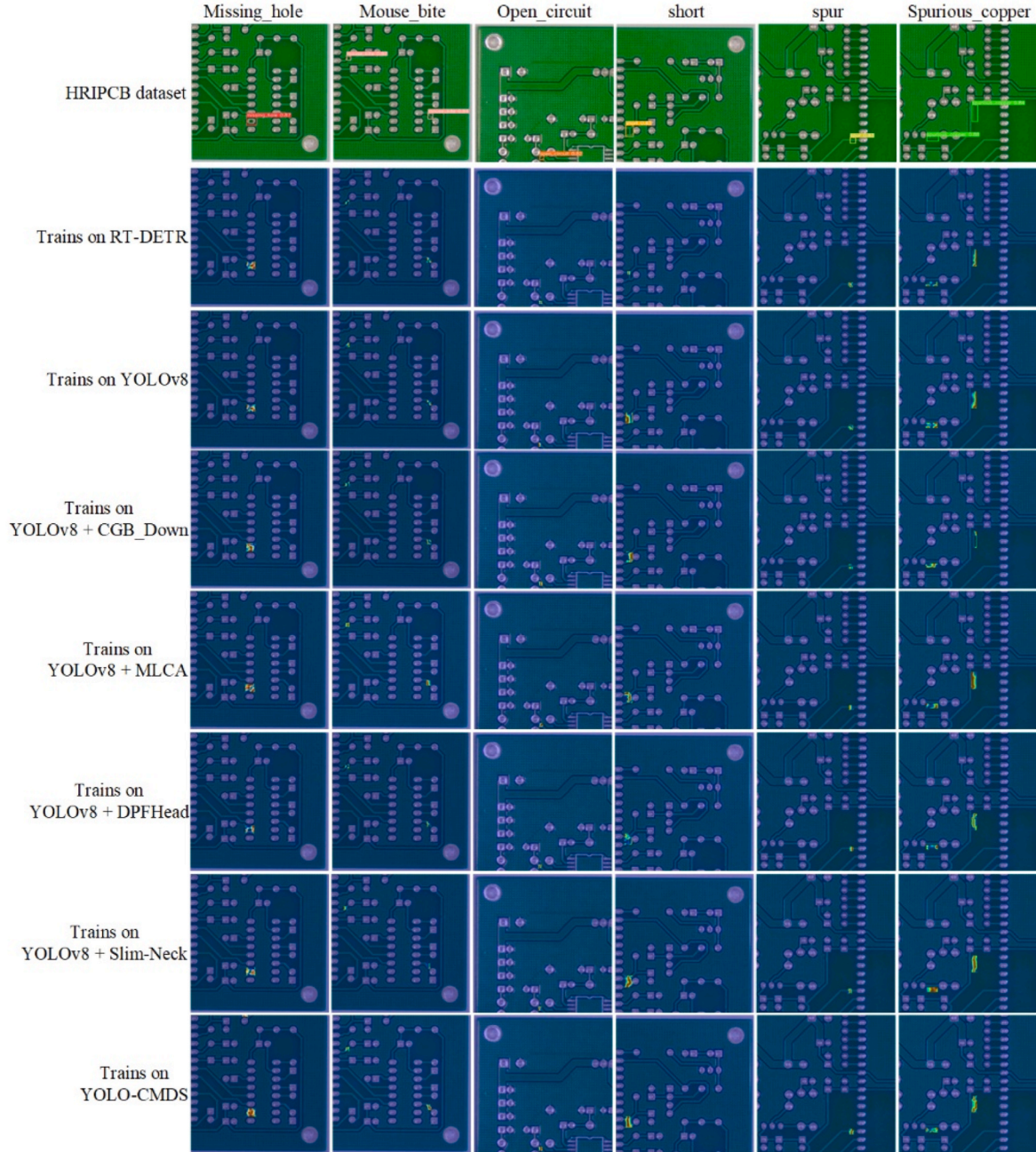
follows: YOLOv8 as model (1), YOLOv8 + CGB\_Down as model (2), YOLOv8 + CGB\_Down + MLCA as model (3), YOLOv8 + CGB\_Down + MLCA + DPFHead as model (4), and YOLOv8 + CGB\_Down + MLCA + DPFHead + Slim-Neck as YOLO-CMDS (model 5) (Note: "+" indicates "addition"). The results are presented in Table 4.

Comparing models (1) and (2), replacing the original YOLOv8 backbone resulted in a slight mAP decrease from 98.73% to 98.65%. However, detection accuracy for other defects improved significantly. This improvement is due to the CGB\_Down module, which enhances small defect detection. Comparing models (2) and (3), the mAP increased, showing that the Hybrid Local Channel Attention Mechanism (MLCA) improved feature learning and detection performance. From

**Table 5**

Independent module evaluation.

| Model       | CGB_Down | MLCA | DPFHead | Slim-Neck | mAP0.5 | GFLOPs | FPS  |
|-------------|----------|------|---------|-----------|--------|--------|------|
| Baseline    | —        | —    | —       | —         | 98.73  | 8.2    | 78.7 |
| + CGB_Down  | ✓        | —    | —       | —         | 98.65  | 8.5    | 76.4 |
| + MLCA      | —        | ✓    | —       | —         | 98.81  | 8.4    | 77.1 |
| + DPFHead   | —        | —    | ✓       | —         | 98.95  | 8.8    | 73.0 |
| + Slim-Neck | —        | —    | —       | ✓         | 98.88  | 8.6    | 74.2 |

**Fig. 9.** Grad-CAM based feature visualization comparison of different module configurations on representative PCB defects.

model (3) to model (4), the mAP increased further, demonstrating that the DPFHead module improved small defect detection and overall performance. Finally, model (5), YOLO-CMDS, showed significant accuracy improvements, with the mAP rising and confirming the effectiveness of the proposed enhancements.

In addition to accuracy metrics, computational efficiency was also

evaluated in the ablation study. As shown in Table 4, the introduction of individual modules leads to a gradual increase in computational complexity, accompanied by a moderate reduction in inference speed. However, these efficiency costs are compensated by consistent improvements in detection accuracy, particularly for small PCB defects. Notably, some intermediate configurations achieve higher inference

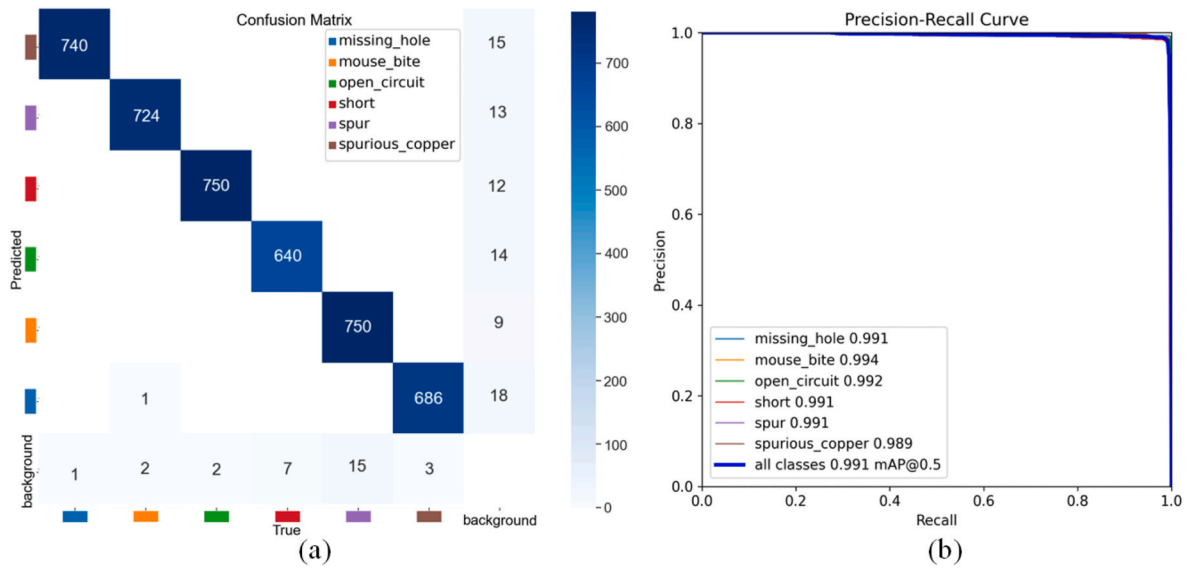


Fig. 10. Display of training results. (a) confusion matrix diagram. (b) Precision-Recall curve.

speed with slightly lower accuracy, indicating a flexible trade-off between performance and efficiency. This validates the rationality of the proposed design choices for practical PCB inspection scenarios with different real-time requirements.

To further analyze the independent contribution of each module and examine potential coupling effects, we conducted additional experiments where each module was separately integrated into the baseline YOLOv8. The results, shown in Table 5, indicate that each component independently improves detection performance to varying degrees. This demonstrates that the proposed modules are not strongly dependent on each other and contribute complementary improvements to the overall architecture.

YOLO-CMDS shows a marked accuracy improvement over model (1), proving it can accurately detect tiny defects in compact PCB layouts with complex backgrounds. Thus, YOLO-CMDS demonstrated excellent performance in detection tasks.

#### 4.4. Visualization results

Fig. 9 presents Grad-CAM visualization results for different model configurations, including the baseline YOLOv8, YOLOv8 with individual modules (CGB\_Down, MLCA, DPFHead, Slim-Neck), and the final YOLO-CMDS model. Six representative defect categories from the HRIPCB dataset are selected. Compared with the baseline, each individual module shows more concentrated activation on defect regions and reduced background interference. Among them, CGB\_Down enhances contextual awareness, MLCA strengthens channel discrimination, DPFHead improves localization precision, and Slim-Neck refines multi-scale feature fusion. The final YOLO-CMDS exhibits the most focused and stable activation patterns across all defect types, further validating the complementary contributions of the proposed modules.

Fig. 10(a) presents the confusion matrix for evaluating object recognition performance. In the confusion matrix, rows and columns represent actual and predicted classes, with diagonal values indicating correct predictions. Fig. 10(b) shows the precision-recall (PR) curve. Precision and recall are trade-off metrics; high precision and recall indicate strong model performance. A PR curve closer to the top-right corner reflects better model robustness and performance. These results demonstrate that YOLO-CMDS performs well in defect detection tasks.

## 5. Conclusion

This paper introduces YOLO-CMDS, a novel model for PCB defect detection. The model improves small defect feature extraction in complex PCB backgrounds through the CGB\_Down module. The Hybrid Local Channel Attention mechanism (MLCA) boosts detection accuracy with minimal increase in parameters. The DPFHead detection head further enhances detection performance for medium and small defects. The Slim-Neck module facilitates feature fusion between the backbone and detection head, improving small defect detection and overall performance.

Extensive experiments show that YOLO-CMDS outperforms state-of-the-art models in both accuracy and efficiency, with an accuracy improvement of up to 99.13% over YOLOv8. Ablation experiments confirm that each design strategy enhances performance, highlighting YOLO-CMDS's ability to detect small defects in complex PCB layouts.

In practical industrial PCB inspection scenarios, real-time performance and deployment efficiency are critical considerations. Although this work primarily focuses on algorithm-level optimization and GPU-based evaluation, the proposed YOLO-CMDS has been validated in real-world industrial applications through system-level deployment in collaboration with industry partners. Due to platform-specific implementation details and confidentiality constraints, hardware-level evaluations on FPGA or ASIC platforms are not included in this study. Future work will explore hardware-oriented optimization and embedded deployment of the proposed model.

#### CRedit authorship contribution statement

Junpeng Xu: Methodology, Investigation, Writing – review & editing. Lei Shi: Methodology, Investigation, Writing original draft. Wenqiang Yang: Equipment, Conceptualization. Xiangbo Zhu: Equipment, Conceptualization. Jin Li: Resources, Equipment.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The publicly available dataset analyzed in the current study is available at <https://robotics.pkusz.edu.cn/resources/dataset/>.

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